

REGULAR PAPER

Identification and mechanistic investigation of ellagitannins from *Osbeckia octandra* that attenuate liver fibrosis via the TGF- β /SMAD signaling pathway

Sudarma Bogahawaththa,^{1,2,3} Tomoaki Kawamura,² Udari Bandaranayake,^{3,4} Tomoaki Hirakawa,^{1,2} Goki Yamada,^{1,2} Hana Ishino,⁵ Tsuzumi Hirohashi,⁵ Shin-ichi Kawaguchi,^{1,6} Kavindra K. Wijesundera,⁷ Missaka P.B. Wijayagunawardane,³ Kanji Ishimaru,⁵ Suranga P. Kodithuwakku,^{3,8,*} and Tadayuki Tsujita^{1,2,*}

¹The United Graduate School of Agricultural Sciences, Kagoshima University, Kagoshima, 890-0065, Japan;

²Laboratory of Biochemistry, Department of Advanced Lifesciences and Food Chemistry, Faculty of Agriculture, Saga University, Saga, 840-8502, Japan; ³Department of Animal Science, Faculty of Agriculture, University of Peradeniya, Peradeniya, 20400, Sri Lanka; ⁴Department of Functional Material Science, Graduate School of Science and Engineering, Saitama University, Shimo-Okubo 225, Sakura-ku, Saitama, 338-8570, Japan;

⁵Department of Biological Resource Sciences, Faculty of Agriculture, Saga University, 1 Honjo, Saga, 840-8502, Japan; ⁶Center for Education and Research in Agricultural Innovation, Faculty of Agriculture, Saga University, Karatsu, Saga, 847-0021, Japan; ⁷Department of Veterinary Pathobiology, Faculty of Veterinary Medicine and Animal Science, University of Peradeniya, Peradeniya, 20400, Sri Lanka; and ⁸ERA Chair COMBIVET, Institute of

veterinary Medicine and Animal Science, Estonian University of Life Science, Tartu, 51014, Estonia

*Correspondence: Suranga P. Kodithuwakku, surangap@agri.pdn.ac.lk; Tadayuki Tsujita, tada@cc.saga-u.ac.jp

ABSTRACT

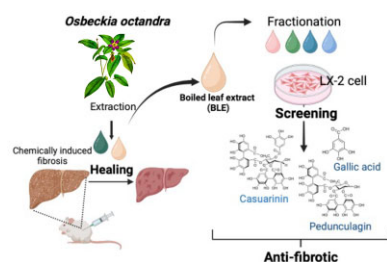
Fibrosis is a major problem in chronic liver disease with limited treatment options due to its complex nature. Herbal medicines are often used as an alternative. The aim of this study was to investigate the therapeutic potential of *Osbeckia octandra* and to identify its active compounds and regulatory pathways. The effects of crude leaf suspension and boiled leaf extract were investigated in an animal model, and the extract was found to be the more effective treatment. Three major bioactive compounds, pedunculagin, casuarinin, and gallic acid, were isolated from the extract using the hepatic stellate cell line, LX-2-based antifibrotic effect evaluation system. The results showed that all these compounds ameliorated LX-2 in fibrotic state. This inhibitory mechanism was confirmed through the TGF- β /SMAD signaling pathway. Collectively, the presence of these compounds in *O. octandra* suggests its potential as a treatment for liver fibrosis.

Received: 20 June 2023; Accepted: 4 August 2023

© The Author(s) 2023. Published by Oxford University Press on behalf of Japan Society for Bioscience, Biotechnology, and Agrochemistry.

All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

Graphical Abstract



LX-2 antifibrotic screening system found that *Osbeckia octandra* boiled leaf extract contains: pedunculagin, casuarinin, and gallic acid as the main therapeutic agents for liver fibrosis.

Keywords: *Osbeckia octandra*, Liver fibrosis, Antifibrotic, LX-2 cell, TGF- β /SMAD

Liver fibrosis is a wound-healing response to an acute or chronic liver injury, which can be caused by various factors such as drug-induced hepatitis, alcoholic and nonalcoholic hepatitis, and hepatitis B and C virus infections (Golabi et al. 2022). The progression of this liver fibrosis is a critical stage and it leads to more severe stages known as liver cirrhosis and hepatocellular carcinoma (HCC) (Murase et al. 2003; Chen et al. 2019). The World Health Organization statistics, cirrhosis affects 1% of the world population, and HCC becomes to the fifth most common cause of cancer (Wong, Ting and Chan 2018). Liver fibrosis and cirrhosis also impose a considerable economic burden to society (Wang et al. 2014). There are several potential drugs, such as doxorubicin (Greupink et al. 2006), metformin (Lavine et al. 2011), and silymarin (Freedman et al. 2011), that have been shown to have antifibrotic effects; however, they have not been able to produce significant therapeutic effects in clinical practice. Liver transplantation is currently the only option for long-term survival in patients with liver disease. However, the procedure is expensive and finding a suitable donor is difficult. Considerable efforts are currently being devoted to research into liver fibrosis, not only with the goal of further understanding the molecular mechanisms that drive fibrosis but also with an equal in focus on establishing effective therapeutic strategies.

The primary mechanism underlying liver fibrosis is the activation of hepatic stellate cells (HSCs), which leads to the excessive accumulation of proteins such as alpha-smooth muscle actin (α -SMA) and collagen in the extracellular matrix (ECM) (Hoffmann et al. 2020). Under the influence of profibrotic markers, (quiescent) HSCs are activated and gradually transformed into myofibroblasts, leading to a metabolic imbalance and large amounts of ECM. As a result, ECM accumulation in the liver can accelerate and exacerbate the pathophysiological process of liver fibrosis (Zhang et al. 2015). Signal transduction pathways play a crucial role in the activation of HSCs and the progression of liver fibrosis. One of the major signaling pathways involved in this process is the transforming growth factor-beta (TGF- β) signaling pathway (Dewidar et al. 2019). Following liver injury, TGF- β is released and activates downstream signaling cascades, including the SMAD-dependent pathway (Gressner, Dooley and Weiskirchen 2005). This pathway triggers the transcription of fibrogenic genes, such as α -SMA and collagen, thereby promoting the synthesis and deposition of ECM components in the liver (Peng et al. 2022). These signaling pathways contribute to the complex molecular processes involved in liver fibrosis, highlighting the importance of understanding and targeting them for potential therapeutic intervention.

Therefore, when investigating new therapeutic strategies to cure liver fibrosis and cirrhosis, the preference is shifting to complementary and alternative medicines, which are either natural products or their derivatives (Sahar S. Atrees 2021). The basis for this preference is their edibility, which is generally accepted as having safe and long-lasting therapeutic potential (Zhang, Sun and Wang 2013). As part of our ongoing projects to explore biologically active and unique natural products from herbal plants, we focused on *Osbeckia octandra*, which has been used as a traditional medicine in Sri Lanka for the treatment of jaundice and viral hepatitis. Previous pharmacological studies on *O. octandra* have shown that extracts of this plant have anti-oxidative, anti-inflammatory, and anti-cancer properties (Prasadani et al. 2021). In our bioactivity screening, we found that the aqueous extract of *O. octandra* has prophylactic potential for ameliorating liver fibrosis and cirrhosis (Bogahawaththa et al. 2021). Several bioactive chemical components, gallic acid, protocatechuic acid, kaempferol, and quercetin have been isolated from another species of the same plant family, namely *Osbeckia aspera* (Grayer et al. 2008). However, the exact antifibrotic effects of these compounds remain poorly understood. As for *O. octandra*, no efforts have been made to isolate and identify any specific compounds from this species. Therefore, it is worthwhile to identify the antifibrotic compounds present in the crude extracts and their intervention in the pathways of fibrosis progression.

In this study, our first objective was to evaluate the therapeutic potential of *O. octandra* in a chemically induced fibrosis rat model. We then isolated the key antifibrotic compounds using LX-2, a fibrotic screening model based on human-derived HSCs and investigated their therapeutic effects. Finally, we discovered that three compounds interfered with the TGF- β signaling pathway and its downstream SMAD signaling pathway. Taken together, these findings suggest that suppression of TGF- β signaling by *O. octandra* could potentially return HSCs to their quiescent state, thereby inhibiting fibrogenesis and promoting liver recovery.

Materials and methods

Plant materials

Leaves of *O. octandra* were collected from the garden of the Department of Animal Science, Faculty of Agriculture, University of Peradeniya, Sri Lanka. The plant specimens were authenticated by the Curator of the Royal Botanical Gardens in Peradeniya, Sri

Lanka, and a voucher specimen (*O. octandra* specimen No. UB 89) was deposited in the National Herbarium.

Leaf extracts preparation

Fifty grams of fresh of *O. octandra* leaves were washed with distilled water (DW) and air dried at room temperature. The air-dried leaves were freeze-dried in a freeze-drier, Alpha 1-4 LD plus (Christ, Osterode am Harz, Germany), ground and sieved to a size of 0.05 mm. Samples were stored in a refrigerator at 4 °C until further use. Boiled leaf extract (BLE) was prepared by mixing 30 g of freeze-dried powder with 480 mL of DW and boiling for 20 min. Powdered extracts were stored at 4 °C until further use. Freeze-dried powder of each extract was dissolved in 500 mg/mL of DW to prepare a stock solution just before the administration.

Experimental animals

Six-week-old, male Wistar rats (220–240 g) were housed at the vivarium of the Faculty of Medicine, University of Peradeniya, Sri Lanka, and reared under the standard conditions (22 ± 3 °C with a 12-h light-dark cycle). A total of 10 rats were randomly assigned as the normal control group and 20 rats were injected with thioacetamide (TAA) (Merck, Darmstadt, Germany) dissolved in physiological saline (154 mM NaCl) at a dose of 100 mg/kg body weight (BW), intraperitoneally (i.p.), twice a week, for 5 weeks to induce fibrosis. Five rats from the control group 1 (CON1) and five rats from the TAA induced group (TAA) were sacrificed at the end of 5 weeks. The remaining 5 rats of the initial control group were kept as the control group 2 (CON2) and the remaining 15 rats of the TAA treated group were divided into three groups for the next 5 weeks of the study. Induced not treated group (INT) was given DW after the week 5. They were kept observing the natural regeneration of hepatocytes after chronic liver damage. The CLS only group was received CLS (500 mg/kg BW, orally, twice a week) and the BLE only group received BLE (500 mg/kg BW, orally, twice a week). All rats were euthanized under isoflurane anesthesia 10 weeks after the commencement of treatments and blood and liver samples were collected for analysis. Experimental procedures were conducted in accordance with the Institutional (Faculty ethic committee, Faculty of Veterinary Medicine and Animal Science, University of Peradeniya, Sri Lanka) Animal Ethics Guidelines for the conduct of Animal Experiments and Animal Care (Certificate No. VER-16-001).

Body and liver weights assessment

The BWs of the rats were measured at the beginning of the study and the final BWs were measured before the animals were sacrificed. Liver weights were measured after sacrifice.

Serum collection and blood chemistry

Serum was separated from the collected blood and stored at -20 °C. Samples were analyzed for aspartate aminotransferase (AST) and alanine aminotransferase (ALT) using a 3000 Evolution (Biochemical Systems International, Arezzo, Italy) according to the manufacturer's instructions.

Tissue preparation and histopathology

Collected liver samples were fixed in 4% paraformaldehyde in phosphate-buffered saline (PBS) (135 mM NaCl, 10 mM phos-

phate, 2.7 mM KCl, pH 7.4), embedded, and sectioned using a microtome and mounted on glass slides. Liver sections were stained with hematoxylin and eosin (H&E) for histopathology and Masson's trichrome (MT) for collagen deposition (Jaeschke, Farhood and Smith 1990).

Fractionation of BLE

BLE was prepared by mixing 300 g of freeze-dried powder with 2.4 L of DW and boiling for 20 min. The supernatant was removed, and the solvent was concentrated *in vacuo* and passed through a column of DIAION HP20SS (Mitsubishi Chemical, Tokyo, Japan), which was eluted with DW containing increasing amounts of methanol.

Thin layer chromatography

Thin layer chromatography (TLC) was performed on precoated 0.2 mm thick Kieselgel 60 F254 plates (Merck) were developed with chloroform/methanol/water (8:5:1, v/v/v), or toluene/ethyl formate/formic acid (1:7:1, v/v/v). Spots were detected under UV illumination after spraying with 2% ethanolic FeCl_3 or 5% sulfuric acid and heating.

HPLC analysis

High performance liquid chromatography (HPLC) was used to identify the peaks for BLE, each fraction (F1–F4) and isolated compounds by using an LC-20AT HPLC system (Shimadzu, Kyoto, Japan) equipped with a COSMOSIL Cholesterol, 4.6 mm I.D. $\times$ 250 mm, reversed-phase column (Nacalai Tesque, Kyoto, Japan). The mobile phase was 0.1% formic acid/ H_2O – CH_3CN (10%–80% in 30 min with linear gradient elution) at a flow rate of 0.6 mL/min, temperature 40 °C, and the absorbance of the effluent solution was measured from 200 to 700 nm using an SPD-M20A photodiode array detector (Shimadzu).

Cell culture

The human hepatic stellate cell line LX-2 was donated by Dr. Scott L. Friedman (Mount Sinai School of Medicine, New York, NY, USA) and was cultured in high glucose Dulbecco's modified eagles medium (DMEM) (Fujifilm Wako, Osaka, Japan) supplemented with 10% fetal bovine serum (FBS) (Equitec-Bio, Kerrville, TX, USA), 100 U/mL penicillin, and 100 $\mu\text{g/mL}$ streptomycin (Nacalai Tesque) in a humidified atmosphere containing 5% CO_2 at 37 °C.

Cytotoxicity and cell viability assay

LX-2 cells (4.0×10^4 cells/well) were seeded in a 96-well plate and incubated for 16 h. The medium was replaced with fresh medium containing *O. octandra* fractions or isolated compounds. The cells were then stimulated for 24 h and to analyze cell viability, the medium was replaced with fresh medium containing 1/10 (v/v) of CCK-8 assay reagents (Dojindo, Kumamoto, Japan) and incubated for 1 h. After incubation, the OD_{450} of the CCK-8 reagents was measured using the SpectraMax i3x (Molecular Devices, San Jose, CA). Cell viability was expressed as a percentage of cells stimulated with 1% dimethyl sulfoxide (DMSO) and used as a negative control.

Induction strategy to identify antifibrotic effect

LX-2 cells were activated with 10% FBS for 24 h to induce fibrosis and then different fractions were treated for 24 h. To compare the treatment effect, three controls were maintained: to observe the activation of LX-2, fibrosis induced group (control F); treated with 10% FBS only, to observe the restoration from active to quiescent state, gradually recovered group (control R); treated with 2% FBS and to observe the clinically available drug, treatment positive group (control T); treated with 5 μ M of doxorubicin (Toronto Research Chemicals, Toronto, ON, Canada) for 24 h (Figure 4b). The antifibrotic effect of each fraction was determined by examining the fibrotic markers; TGF- β , α -SMA, COL1A1, CD44, SMAD2/3, and SMAD7.

Isolation and identification of major bioactive compounds

Based on the results of the antifibrotic study for F1–F4, we selected F3 and F4 for further purification. To purify these fractions, we used a stepwise elution process with DW and methanol using Sephadex LH20 (Sigma, St. Louis, MO, USA) and TOYOPEARL HW-40F (Tosoh, Tokyo, Japan) as the purification medium. To identify the peaks of the purified compounds, nuclear magnetic resonance (NMR) spectra were obtained using an AVANCE NEO 400 spectrometer (Bruker, Billerica, MA, USA). DMSO- d_6 was used as a solvent for casuarinin (CSU) and gallic acid (GA), methanol- d_4 for pedunculagin (PDN). For further confirmation of PDN and CSU, high resolution electrospray Ionization-Mass Spectrometry (HRESI-MS) analysis was done for the confirmation of PDN and CSU with high resolution mass spectrometer, microTOF II (Bruker). The mass spectrometer conditions were capillary voltage 3.5 kV, desolvation temperature 200 °C, gas flow 4 L/min, and gas pressure 0.3 bar. Nitrogen was used as drying and collision gas, respectively. The data were analyzed using the Data Analysis 4.1 software (Bruker).

Immunoblot analysis

The LX-2 cells (5.0×10^5 cells/well) were seeded in a 12-well plate and cultured in 10% FBS for 24 h. The medium was then replaced with a fresh medium containing different concentrations of *O. octandra* fractions or isolated compounds and stimulated for 24 h. After stimulation, the cells were lysed with sodium dodecyl sulphate (SDS) sample buffer (125 mM Tris-HCl, pH 6.8, 100 mM DTT, 2% [w/v] SDS, 5% [w/v] sucrose, 0.002% [w/v] bromophenol blue), centrifuged at 13000 rpm for 5 min, and boiled at 90 °C for 5 min. The samples were subjected to SDS-polyacrylamide gel electrophoresis and transferred to polyvinylidene fluoride membranes (Fujifilm Wako). The membranes were blocked and incubated with primary antibodies against α -SMA, TGF- β , COL1A1, CD44, and SMAD2/3 (Cell Signaling Technology, Danvers, MA, USA), or β -actin (Fujifilm Wako). The membranes were then washed and incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies. Signals were detected with ImmunoStar LD (Fujifilm Wako) using Lumino Graph I (ATTO, Tokyo, Japan).

RNA isolation and quantitative PCR.

Total RNA from the mouse liver and the cells was extracted using Sepasol RNA I Super G (Nacalai Tesque) according to the manufacturer's instructions. The extracted RNA was then reverse

Table 1. Primers used for the mRNA expression analysis

Gene	Sequence
Rat-Tnf- α	F ^a ; 5'-GGCTGCCCGGACTATGTG-3' R ^b ; 5'-CTCCTGGTATGAAGTGGCAAATC-3'
Rat-Tgf- β	F; 5'-GAGGTGACCTGG GCACCAT-3' R; 5'-GGCCATGAGGAGCAGGAA-3'
Rat- α -Sma	F; 5'-GACCTGAAGTATCCGATAGAACA-3' R; 5'-CACGCGAAGCTCGTTATAGAAG-3'
Rat-18S rRNA	F; 5'-GTAACCCGTTGAACCCCAATT-3' R; 5'-CCATCCAATCGGTAGTAGCG-3'
Human-TGF- β	F; 5'-TACCTGAACCCGTTGTCTCTC-3' R; 5'-GTTGCTGAGGTATCGCCAGGAA-3'
Human- α -SMA	F; 5'-GCTGAAGTATCCGATAGAACACG-3' R; 5'-GGTCTCAAACATAATCTGGGTGA-3'
Human-CD44	F; 5'-GCTTCCAGAGTTACGCCCTGA-3' R; 5'-AACCCTTGCAACATTGCCTGA-3'
Human-COL1A1	F; 5'-GTACTGGATTGACCCGAACC-3' R; 5'-CGCCATACTCGAACTGGAAT-3'
Human-SMAD2	F; 5'-CCGACACACCGAGATCCTAAC-3' R; 5'-AGGAGGTGGCGTTCTTGGAAT-3'
Human-SMAD3	F; 5'-CCATCTCTACTACGAGCTGAA-3' R; 5'-CACTGCTGATTCTGTGTGAC-3'
Human-SMAD7	F; 5'-CTCGGACAGCTCAATTCGGA-3' R; 5'-CAGTGTGCGGACTTGATGA-3'
Human-18S rRNA	F; 5'-GTAACCCGTTGAACCCCAATT-3' R; 5'-CCATCCAATCGGTAGTAGCG-3'

^a F; forward primer. ^b R; Reverse primer.

transcribed using the FastGene cDNA Synthesis 5x ReadyMix (Nippon Genetics, Tokyo, Japan). The resulting cDNA was used as a template for quantitative polymerase chain reaction (qPCR) using the KAPA SYBR Fast qPCR Master Mix (2 $\times$) Kit (Kapa Bioscience, Potters Bar, UK) with the Mx3005P qPCR system (Agilent Technologies, Santa Clara, CA, USA). The primers used for this study are listed in Table 1. The 18S rRNA expression was used as an internal control to normalize the mRNA expression data (Blanquicett et al. 2002; Lardizábal et al. 2012).

Immunohistochemistry

Immunohistochemistry for TGF- β in liver sections was performed according to the protocol described herein. Briefly, dewaxed and rehydrated tissue sections were blocked and then incubated with TGF- β antibody (Abcam, Cambridge, UK) at 4 °C for 24 h. They were incubated with Dual Link System-HRP labeled secondary antibody (Agilent Technologies) for 30 min at room temperature and visualizing according to the manufacturer's recommendations. The slides were then mounted with a coverslip. Photographs were taken using an IX73 inverted microscope, and digital images were captured using DP27, and analyzed with CellSens software (Olympus, Tokyo, Japan).

Immunofluorescence for CD44 was performed according to a previously published protocol (Ooki and Hatakeyama 2021). Briefly, LX-2 cells were cultured and treated in chamber slides and the cells were prefixed with formaldehyde and permeabilized with Triton X-100. The cells were then incubated with anti-CD44 antibodies (Cell Signaling Technology) overnight at 4 °C. The cells were then washed three times with 1 $\times$ PBS.

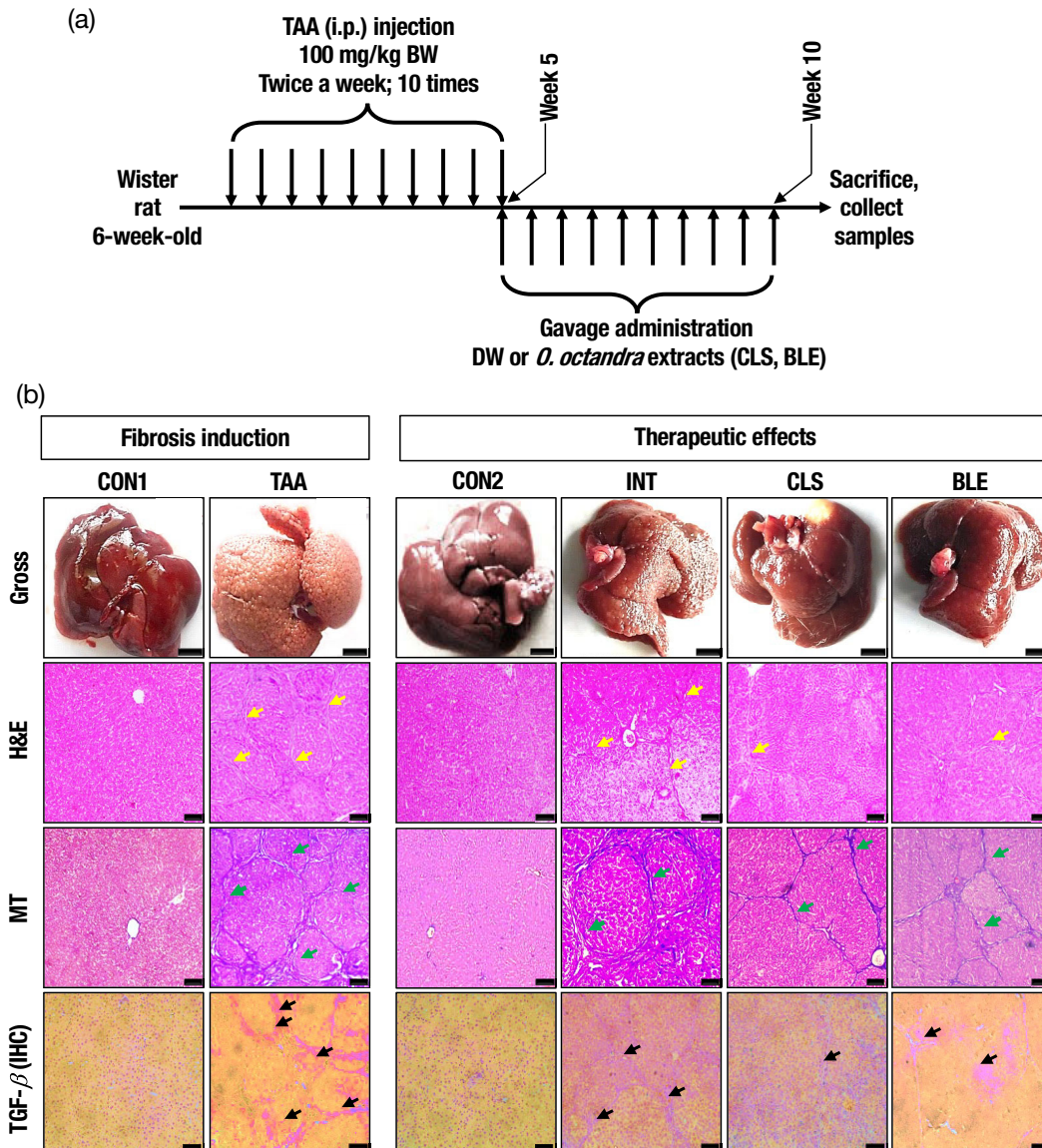


Figure 1. CLS and BLE treatments contribute to the reversal of TAA-induced fibrosis in rats. (a) Experimental design of an animal study to evaluate the therapeutic effect of CLS and BLE after induction of fibrosis in rats. (b) Treatment of CLS and BLE improves the restoration of liver. First two columns show the induction of fibrosis in TAA group compared to CON1. Next four columns show the therapeutic effect of *O. octandra* compared to INT group. First row shows gross pathological observations. Scale bar = 10 mm. Second row shows improvement in liver architecture (H&E staining). Third row shows the reduction of the collagen deposition (MT staining). Fourth row shows the ameliorated expression of TGF- β is with the treatment of CLS and BLE. Scale bar = 100 μ m. Yellow arrows; fibrous bridges, green arrows; collagen deposition and black arrows; TGF- β expression.

DyLight 488 conjugated goat anti-mouse-IgG (Bio-Rad, Hercules, CA, USA) was added as a secondary antibody and incubated for 1 h at room temperature. This was followed by co-staining with PureBlu DAPI nuclear staining dye (Bio-Rad). Finally, images were captured using a DMi8 inverted microscope and processed using the THUNDER imaging system (Leica Microsystems, Wetzlar, Germany).

Statistical analyses

Statistical analyses were performed using one-way analysis of variance (ANOVA) with a Dunnett's test. Data are expressed as mean $\pm$ standard error of the mean (SEM) or mean $\pm$ standard deviation (SD).

Results

Therapeutic effect of *O. octandra* leaf extracts in TAA-induced liver fibrosis

To evaluate the therapeutic effect, first fibrosis was induced by intraperitoneal (i.p.) injection of TAA for 5 weeks in a rat model as described in Figure 1a. In this study, TAA-induced liver injury was assessed by gross-pathology, histopathology, and immuno-histochemistry. At the end of the week 5, the liver surfaces of the control (CON1) group were normal with a dark red in color and no nodules with smooth appearance. The livers of TAA-injected rats were pale in color with scattered nodules on the surface, confirming the induction of fibrosis. After oral administration of *O. octandra*: compared to the INT livers, the progression of

granular nature was reduced with the treatment of CLS and BLE (Figure 1b Gross).

To examine the hepatic architecture of the livers, H&E staining was performed. At the end of week 5, the CON1 group showed a regular hepatic architecture with no pathological abnormalities. The fibrosis induced TAA group showed loss of normal architecture with sinusoidal dilation (yellow arrows indicate the development of fibrous bundles around the lobules). At the end of treatment, fibrous bridges were still prominent in the INT group. However, compared to the INT livers, CLS and BLE treated livers showed comparatively less fibrous bridging (Figure 1b H&E).

To evaluate the degree of fibrosis in liver samples, collagen fibers in liver tissue were stained with MT. Compared with the CON1, TAA-injections for 5 weeks showed the initiation of deposition of collagen fibers (green arrows). Furthermore, the INT group still showed collagen deposition and incomplete fibrocepa. However, treatment with CLS and BLE for 5 weeks, ameliorated the deposition of collagen fibers compared to the INT group (Figure 1b MT).

In addition, TGF- β is a key mediator in the establishment of liver fibrosis. Therefore, immunostaining was performed to determine the localization of TGF- β . Compared to the CON1, liver sections from the TAA group showed prominent cytoplasmic staining of TGF- β (black arrows). After 5 weeks, the INT group still showed moderate expression of TGF- β . At the end of *O. octandra* treatment, liver sections from CLS and BLE treated rats showed comparatively lower expression of TGF- β compared to the INT group (Figure 1b TGF- β IHC).

O. octandra leaf extracts improve the biological and biochemical indices of fibrosis progression

The establishment of liver fibrosis and the therapeutic effect of *O. octandra* leaf extracts were observed by monitoring BW gain (the difference between initial and final BW) and the liver index (liver weight/BW, percentiles). With the induction of liver fibrosis, the TAA group showed a significant decrease in BW gain compared to the CON1 group at the end of the fibrosis induction period. After treatment, the CLS and BLE treated groups showed a prominent recovery of BW gain compared to the INT group (Figure 2a). At the same time, the TAA group showed a notable increase in liver index compared to the CON1 group at week 5 indicating the establishment of the fibrosis. Notably, the CLS and BLE treatments showed a significant reduction in liver indices compared to the INT group (Figure 2b).

Induced liver fibrosis and the extent of recovery with *O. octandra* treatment were further evaluated by liver function tests. Serum levels of AST and ALT are markers of liver injury (Ramaiah 2007). Therefore, we assessed the degree of liver injury by determining the levels of serum ALT and AST at weeks 5 and 10. AST and ALT were considerably higher in the TAA treated group than in the CON1 group, indicating the establishment of fibrosis at week 5 (Figure 2c and d). Compared to the start of treatment, the CLS and BLE groups had significantly lower AST levels at the end of week 10, than the INT group (Figure 2c). Moreover, the CLS and BLE groups also had notably lower ALT levels than the INT group (Figure 2d). These results further confirm that induced fibrosis is restored by treatment with *O. octandra*.

O. octandra leaf extracts ameliorate liver fibrosis by mediating key molecular markers of fibrosis

The mRNA expression levels of pro-inflammatory and fibrotic cytokines were examined (Figure 3a–c). Compared with the

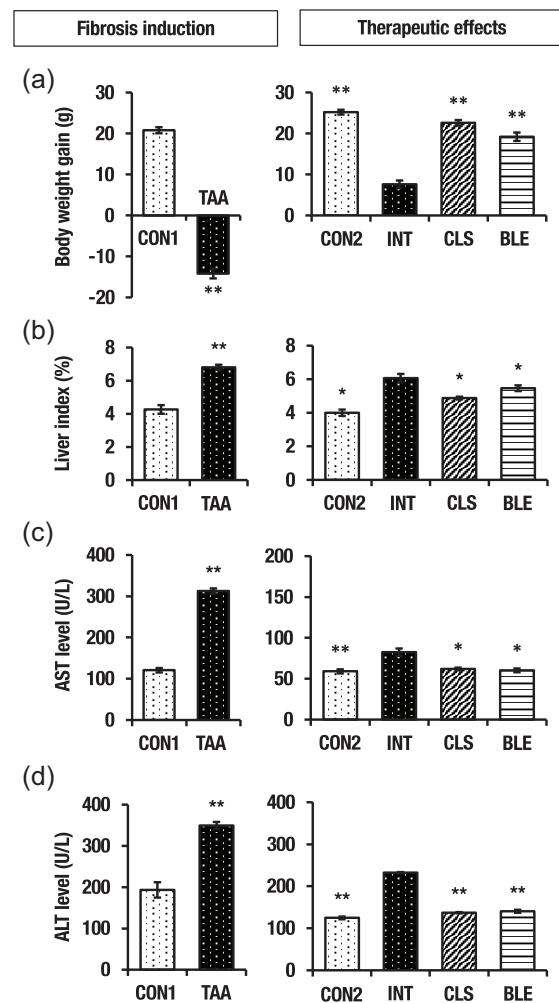


Figure 2. CLS and BLE treatments restore liver condition as measured by body weight, liver weight, and serum parameters. (a) Body weight gain after the fibrosis induction and after treatment with CLS and BLE. Values are in grams of increase/decrease from the starting weight. (b) Liver index (Liver/BW ratio) after induction of fibrosis and after treatment with CLS and BLE. (c) Serum AST levels with the induction of fibrosis and after treatment with CLS and BLE. (d) Serum ALT levels after the induction of fibrosis and after treatment with CLS and BLE. Statistical significance has been indicated in two ways; (i) Disease induction effect during the TAA challenge (CON1 vs. TAA). (ii) Treatment's effect (INT vs. CON2, CLS, and BLE), with one-way ANOVA followed by Dunnett's test. * $P < 0.05$, ** $P < 0.01$. Error bars represent SEM in each group ($n = 5$).

CON1, significantly higher levels of *Tnf- α* , *Tgf- β* , and *α -Sma* mRNA expression were observed after induction of the fibrosis with TAA at week 5, confirming the establishment of fibrosis. To evaluate the therapeutic effects of CLS and BLE, the above markers were also analyzed at the end of week 10 (5 weeks of treatment). Interestingly, the expression levels of *Tnf- α* , *Tgf- β* , and *α -Sma* were clearly downregulated in the CLS and BLE treated groups compared to INT, but higher than in the CON2 group. These results confirm that both CLS and BLE can be used as a therapeutic option for the liver fibrosis.

Bulk fractionation for the identification of antifibrotic compounds in *O. octandra* boiled leaf extract

To identify possible bioactive compounds, which are responsible for the therapeutic effect, BLE was fractionated by column chromatography (DIAION as stationary phase and aqueous methanol

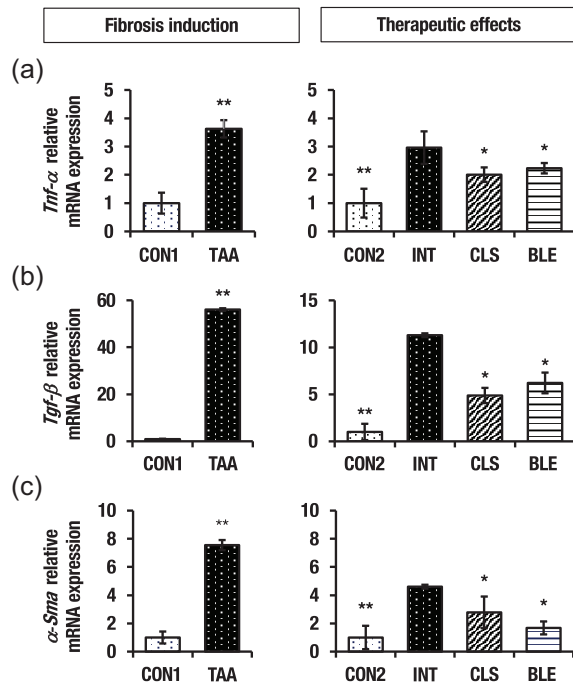


Figure 3. CLS and BLE treatments showed hepatoprotective effects by regulating gene expressions in TAA-induced fibrosis. (a–c) Relative mRNA expression of *Tnf-α* (a), *Tgf-β* (b), and *α-Sma* (c) after induction of fibrosis and therapeutic effect after treatment with CLS and BLE. Expression levels are normalized to 18S rRNA level. Data based on the value from CON1 and CON2 group set as 1.0. The statistical significance has been indicated in two ways: (i) Disease induction effect during the TAA challenge (CON1 vs. TAA). (ii) Treatment effect (INT vs. CON2, CLS, and BLE), with one-way ANOVA followed by Dunnett's test. * $P < 0.05$, ** $P < 0.01$. Error bars represent SEM in each group ($n = 5$).

as mobile phase). At the end of this process, the eluates were separated into four fractions (dry weights for F1–F4 were 26, 8.0, 3.0, and 8.0 g, respectively) (Figure 4a) based on the spots on the TLC plate after spraying anisaldehyde and/or FeCl_3 under UV light.

LX-2 fibrosis model allows identification of the antifibrotic fractions from boiled leaf extract

To evaluate the antifibrotic effect of each fraction, the LX-2 cell line was used. The LX-2 cells show characteristics of primary HSCs *in vivo* and show a quiescent phenotype in growth medium with low serum concentrations (2% FBS) but gets activated in high serum concentrations (10% FBS). To activate quiescent LX-2 cells, they were cultured in 10% FBS for 24 h. This activation mimics the fibrosis in HSCs. Different fractions were then used to treat the cells at different doses for another 24 h (Figure 4b).

To select the best concentration series to study the therapeutic effect of each fraction, cell viability, and IC_{50} values were examined using the CCK-8 assay. As shown in the Figure S1, a cell viability decreased in a dose-dependent manner. Calculated IC_{50} values for F1; 1362, F2; 2016, F3; 890, and F4; 1621 $\mu\text{g/mL}$, respectively (Figure S1b). According to these values, three doses; 1000, 500, and 250 $\mu\text{g/mL}$ were selected to evaluate the antifibrotic effect of each of the fractions. Protein expression of key fibrotic markers was also examined. According to the immunoblotting results (Figure 4c), with respect to the three controls used in this study, the continuous fibrosis induced group (control F) showed significantly higher TGF- β , α -SMA, and collagen 1 $\alpha 1$ (COL1A1) protein expression levels compared to the gradually recovered

group with 2% FBS treatment (control R), confirming the activation of HSCs. Notably, after the treatments, F3 and F4 showed remarkably lower expression levels of fibrotic markers: TGF- β , α -SMA, and COL1A1, compared to F1 and F2 (Figure 4d–f). Furthermore, the strongest antifibrotic effect could be observed in F3 and F4 compared to control R and the positive control (control T) treated with doxorubicin. Based on these results, F3 and F4 were used for further isolation of bioactive compounds responsible for the antifibrotic effect.

PDN, CSU, and GA were identified as compounds from fractions 3 and 4

BLE and fractions (F1–F4) were analyzed by HPLC and the profiles are shown in Figure S2a–e. To identify the major compounds, present in F3 and F4, the fractions were chromatographed, and three compounds were isolated. These isolated compounds were confirmed as pedunculagin (PDN), casuarinin (CSU), and gallic acid (GA) by HRESI-MS (PDN and CSU) and NMR (PDN, CSU, and GA) analyses. The profiles for the isolated compounds (retention times: PDN (13.7 and 16.7 min, for α and β -isomer) (Singh et al. 2016), CSU (18.6 min) (Yamanaka et al. 2008), and GA (7.9 min) (Haikal et al. 2021) are also shown in Figure S2f–h.

PDN

HRESI-MS (positive-ion mode) m/z : 807.0648 [$M + \text{Na}$] + (calculated for $\text{C}_{34}\text{H}_{24}\text{NaO}_{22}$, 807.0657). ^1H NMR (400 MHz, CD_3OD): δ H 3.82 (1H, brd, $J = 11.5$ Hz, H-6b α anomer), 3.88 (1H, brd, $J = 12.6$ Hz, H-6b β anomer), 4.15 (1H, m, H-5 β anomer), 4.58 (1H, m, H-5 α anomer), 4.95 (1H, d, $J = 8.1$ Hz, H-1 β anomer), 5.07–5.12 (4H, unresolved, H-2 α , β anomers and H-4 α , β anomers), 5.25–5.33 (3H, unresolved, H-3 β anomer and H-6a β , α anomers), 5.36 (1H, d, $J = 4.0$ Hz, H-1 α anomer), 5.47 (1H, t, $J = 9.6$ Hz, H-3 glucose α anomer), 6.35, 6.36, 6.49, 6.52, 6.58, 6.60, 6.61 (s, HHDP-H). ^{13}C NMR (125 MHz, CD_3OD): δ C 62.7 (Glu C-6ab), 66.4 (Glu C-5a), 68.5 (Glu C-4a), 69.0 (Glu C-5b), 71.9 (Glu C-4b), 75.0 (Glu C-2a), 75.8 (Glu C-3b), 76.7 (Glu C-3a), 77.3 (Glu C-2b), 90.8 (Glu C-1a), 94.4 (Glu C-1b), 106.3, 106.4, 106.7, 107.2 (HHDP C-3), 113.5, 115.2 (HHDP C-1), 124.6, 124.9, 125.0 (HHDP C-2), 136.1 (HHDP C-5), 143.5, 144.4 (HHDP C-4,6), 167.7, 168.1, 168.3, 168.8, 169.4 (HHDP COO) (Figure S3a).

CSU

HRESI-MS (positive-ion mode) m/z : 959.0877 [$M + \text{Na}$] + (calculated for $\text{C}_{41}\text{H}_{28}\text{NaO}_{26}$, 959.0767). ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ H 3.96 (1H, d, $J = 12.8$ Hz, Glu H-6a), 4.53 (2H, m, Glu H-2, H-6b), 5.15–5.23 (3H, m, Glu H-3, H-4, H-5), 5.37 (1H, br.s, Glu H-1), 6.23, 6.28, 6.56 (each 1H, s, HHDP protons), 6.92 (2H, s, galloyl protons). ^{13}C NMR (125 MHz, $\text{DMSO}-d_6$): δ C 64.3 (Glu C-6), 66.1 (Glu C-1), 68.8 (Glu C-3), 70.0 (Glu C-5), 73.4 (Glu C-4), 75.8 (Glu C-2), 103.6, 105.0, 106.6 (HHDP C-3), 109.2 (galloyl C-2,6), 114.8, 115.8, 116.3, 116.7 (HHDP C-1 and HHDP C-3, for HHDP attached to C-1 glu), 119.1 (galloyl C-1), 119.6, 123.1, 125.4, 126.1 (HHDP C-2), 134.3, 135.0, 136.5, 138.1 (HHDP C-5), 139.4 (galloyl C-4), 143.0, 144.7, 144.8 (x2), 145.2, 145.3, 145.4, 146.0 (HHDP C-4,6), 145.4 (galloyl C-3,5), 163.8 (galloyl COO), 164.9, 168.2, 168.7, 169.4 (HHDP COO) (Figure S3b).

GA

^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ H 6.92 (2H, s, galloyl protons) (Figure S3c).

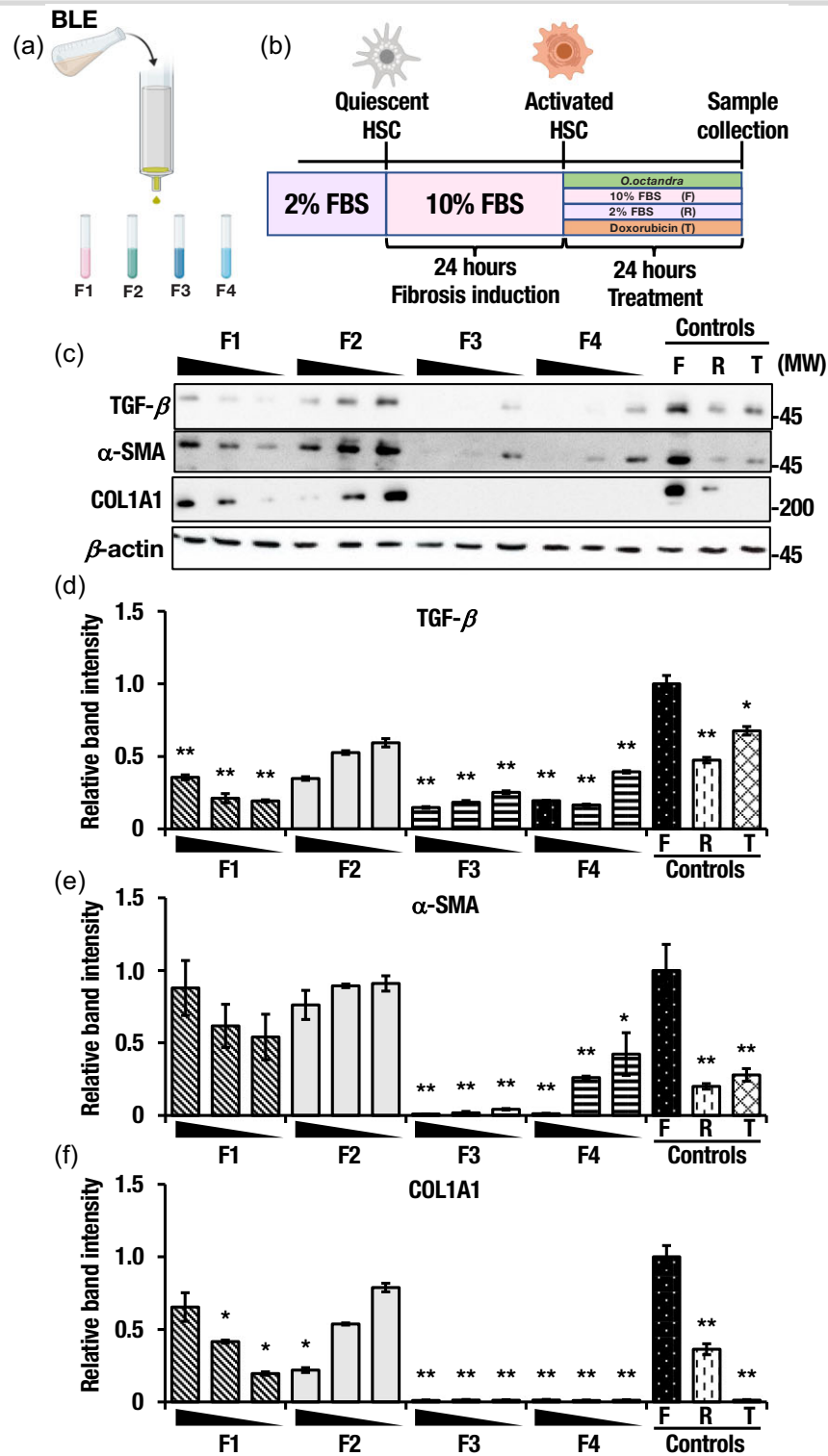


Figure 4. Fraction 3 and 4 showed antifibrotic effect by reducing the expression of fibrotic markers in activated HSCs. (a) Fractionation of the BLE to identify the fraction that contains antifibrotic compounds. (b) Cell culture model established in LX-2 cells to identify the antifibrotic effect. (c) Immunoblot analyses of key fibro genetic markers: activated LX-2 cells were treated with various concentrations (1000, 500, and 250 µg/mL) of different fractions (F1-F4) and fibrosis induced group (F); treated with 10% FBS only, gradually recovered group (R); treated with 2% FBS and treatment positive group (T); treated with 5 µM of doxorubicin for 24 h. Equal loading was assed by probing the blots with antibody against β-actin. (d-f) Relative band intensity of TGF-β (d), α-SMA (e), and COL1A1 (f). Band intensities were measured with image J software. Data based on the value from control F set as 1.0. *P < 0.05, **P < 0.01 (one way ANOVA with Dunnett's test). Each data is mean ± SD (n = 3).

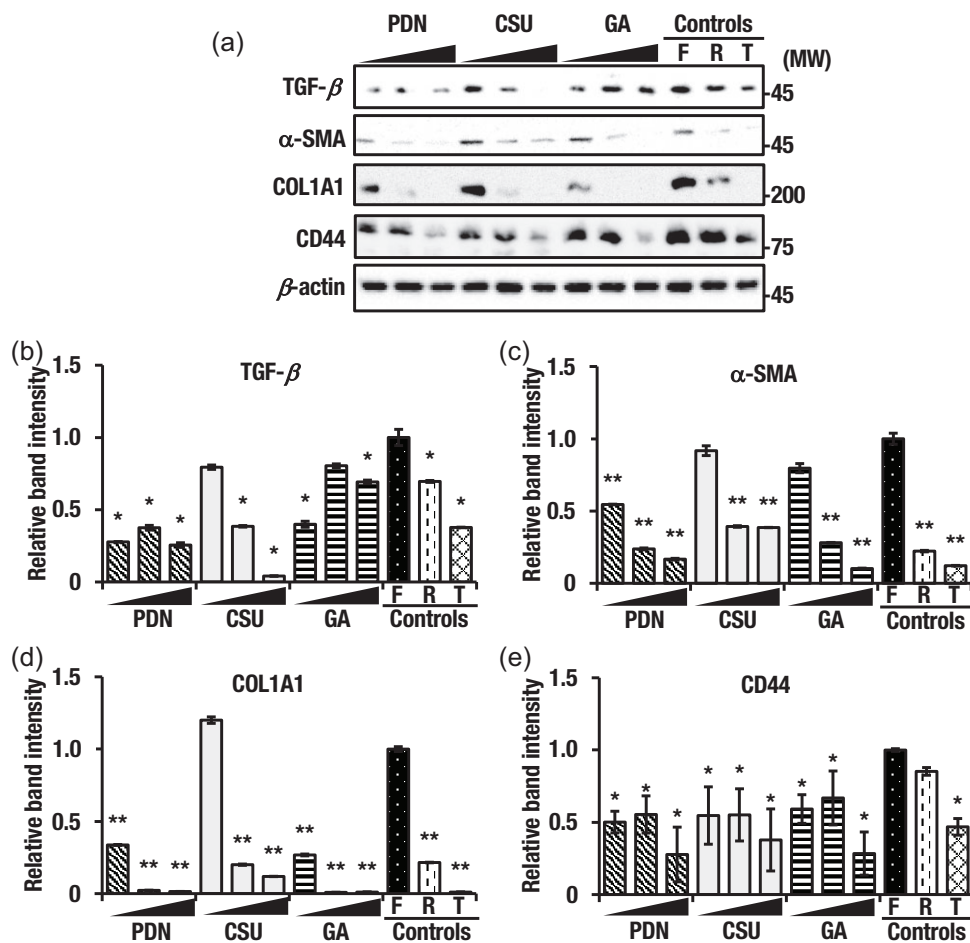


Figure 5. PDN, CSU, and GA mediate fibrosis via regulation of key fibrotic markers. (a) Immunoblot analyses of key fibro genetic markers: activated LX-2 cells were treated with different concentrations (25, 50, and 100 μ M) of different isolated compounds and controls (fibrosis induced group (control F); treated with 10% FBS only, gradually recovered group (control R); treated with 2% FBS and treatment positive group (control T); treated with 5 μ M of doxorubicin for 24 h. Equal loading was assayed by probing the blots with antibody against β -actin. (b–e) Relative band intensity of TGF- β (b), α -SMA (c), COL1A1 (d), and CD44 (e). Band intensities were measured with image J software. Data based on the value from control F set as 1.0. * $P < 0.05$, ** $P < 0.01$ (one way ANOVA with Dunnett's test). Each data is mean $\pm$ SD ($n = 3$).

Restoration of fibrosis is confirmed with key fibrotic markers in activated HSCs by PDN, CSU, and GA

To evaluate the antifibrotic activity of the isolated and identified compounds, previously established cell culture model was used (Figure 4b). Cells were treated at three different doses of 25, 50, and 100 μ M. Subsequently, key fibro-genetic markers were studied as in previously (TGF- β , α -SMA, and COL1A1). In addition, we performed immunoblot analysis of cluster of differentiation 44 (CD44) expression, which is reported as a newly identified fibrotic marker (Figure 5a). Prior to examining PDN, CSU, and GA activity, the expression profiles of four fibrotic markers were compared between the controls F, R, and T in the LX-2 cell model. Control R and T groups showed noticeably lower TGF- β , α -SMA, and COL1A1 expression compared to control F (Figure 5b–d). In the CD44 expression profile, control R showed similar CD44 expression to control F, whereas the treatment positive group control T showed a significant downregulation of CD44 expression (Figure 5e). The TGF- β , α -SMA, and COL1A1 expression was downregulated by PDN, CSU, and GA treatment in a dose-dependent manner compared to the control F (Figure 5b–

d). CD44 expression level was clearly downregulated by PDN, CSU, and GA treatment, with prominent downregulation was observed at the high dose (Figure 5e). These results suggest that PDN, CSU, and GA have great potential for the treatment of liver fibrosis.

To further confirm the fibrotic treatment marker CD44, immunofluorescence studies were performed. A medium dose (50 μ M) was selected to study the effect of each isolated compound on CD44 expression (Figure 6). Control F showed significantly higher expression of CD44 in the cytoplasm compared to both control R and T. Treatment with PDN, CSU, and GA resulted in remarkably lower cytoplasmic expression of CD44 compared to the control F, visually demonstrating the therapeutic effect and confirming the immunoblot results.

To understand the effect of these isolated compounds on the gene expressions of the above fibrogenetic markers, mRNA expressions were also studied (Figure 7a–d). Consistent with the immunoblot results, PDN, CSU, and GA treatment at high and medium doses showed significant downregulation of TGF- β , α -SMA, and CD44 compared to the control F (Figure 7a, b, d). In addition, COL1A1 expression was downregulated in a

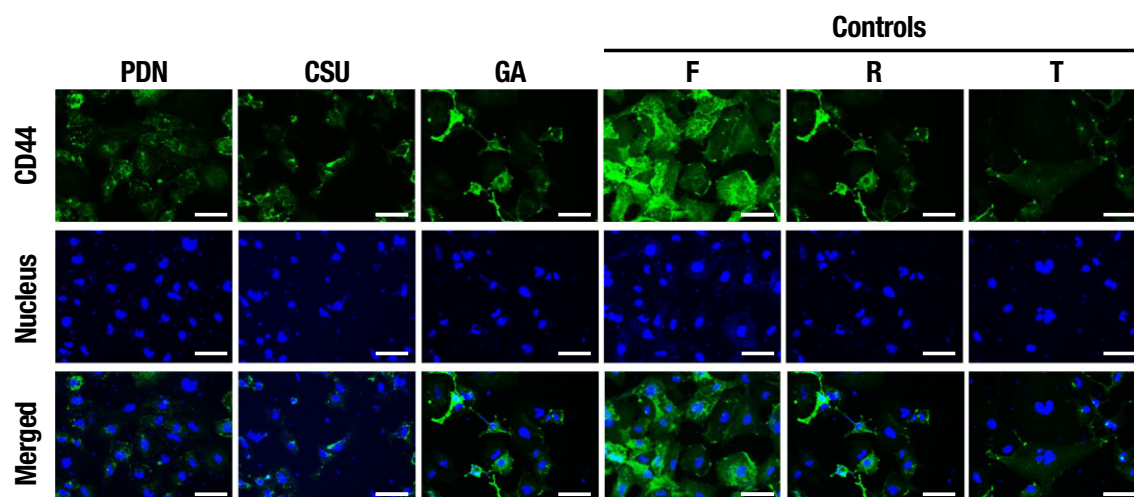


Figure 6. PDN, CSU, and GA mediate fibrosis via down regulation of CD44 expression. Immunofluorescence analyses of CD44: activated LX-2 cells were treated with 50 μ M of different isolated compounds and controls (fibrosis induced group (control F); treated with 10% FBS only, gradually recovered group (control R); treated with 2% FBS and treatment positive group (control T); treated with 5 μ M of doxorubicin for 24 h in activated LX-2 cells. In the first row, CD44 expression is shown in green. In the second row, nuclear staining with DAPI shown in blue. In the third row, both CD44 and DAPI merged images are shown. All images were taken at 200 $\times$ magnification. Scale bar = 50 μ m.

dose-dependent manner in all three treatments (Figure 7c). Notably both the control R and T showed significant downregulation of TGF- β , α -SMA, and COL1A1 compared to the control F. Taken together, PDN, CSU, and GA showed a significant therapeutic effect in activated LX-2 cells.

PDN, CSU, and GA show antifibrotic effects through TGF- β /SMAD signaling pathway

In the antifibrotic study, we and others have reported the significant downregulation of TGF- β at both protein and mRNA expression levels (Varga and Pasche 2008). Therefore, we further investigated the downstream signaling pathway of TGF- β /SMAD signaling, which is a key pathway leading to liver fibrosis (Figure 8a). As shown in Figure 8b and c, significant downregulation of both SMAD2 and SMAD3 was observed with PDN treatment in a dose-dependent manner. CSU and GA showed significant downregulation at high and medium doses compared to the control F. In addition, control T also showed significantly low expression of SMAD2 and SMAD3 proteins compared to the control F.

The mRNA expressions also clearly showed that PDN, CSU, and GA downregulated SMAD2 (Figure 9a) and SMAD3 (Figure 9b), to mediate fibrosis, which is negatively regulated by SMAD7, an inhibitory protein for TGF- β signaling. We then examined the mRNA expression of SMAD7. As shown in Figure 9c, SMAD7 is clearly increased in response to treatment of PDN, CSU, and GA compared to the control F. Taken together, PDN, CSU, and GA showed a potential therapeutic effect via intervention of TGF- β /SMAD signaling in liver fibrosis.

Discussion

Fibrosis is the accumulation of excessive fibrous tissue deposition that leads to tissue hardening and impaired those function (Georges et al. 2007). Chronic inflammation accompanies fibrosis and drives its progression. *O. octandra* plant extracts are being investigated as potential treatments for liver fibrosis and inflam-

mation. Previously, we and others have shown that *O. octandra* crude extract has a prophylactic effect curing on liver disease (Thabrew, Joice and Rajatissa 1987; Bogahawaththa et al. 2021). It is also equally important to study the therapeutic effect because it is more important to find the treatment effect after the establishment of liver fibrosis. In this study, as the first step, we conducted an animal study to investigate the therapeutic effect of *O. octandra*. Here before treatment, we established fibrosis in the rat model and after confirming the establishment of fibrosis, we treated *O. octandra* and examined the recovery of fibrosis.

Repeated injections of TAA in rats lead to the establishment of fibrosis, resulting in various effects on their physiology (Wong et al. 2012). TAA administration reduces appetite, leading to a decrease in caloric intake and subsequent suppression of BW gain (Rhiauani et al. 2008). Concurrently, TAA causes severe liver damage characterized by hepatocyte swelling, glycogen accumulation, and lipid droplet formation (Dwivedi and Jena 2018). These changes may further induce increased protein synthesis contributing to elevated liver indices (Lebeaupin et al. 2018). Additionally, AST and ALT activities in the TAA-treated group can be considered as an indicator of liver parenchymal cell damage. Similar findings have been reported in previous studies of TAA-induced liver injury, confirming the establishment of fibrosis (El Awdan et al. 2019; Jantararussamee et al. 2021). As a hepatotoxicant, TAA induces the expression of tumor necrosis factor- α , creating a cytotoxic environment that triggers chronic inflammation and progressive liver fibrosis (Yang and Seki 2015). Another central regulator in chronic liver disease is TGF- β , which contributes to all stages of disease progression by promoting hepatocyte destruction and activating HSCs and fibroblasts (Dewidar et al. 2019). Activation of HSCs can be identified by the expression of α -SMA, a marker of smooth muscle cell differentiation (Cao et al. 2021). Previous studies have also observed elevated inflammatory cytokines and liver damage in TAA-treated rats (Wijesundera et al. 2015; Abdel-Rahman et al. 2021).

To study the natural regeneration of the liver after removal of the TAA injection, the INT group was included in the study.

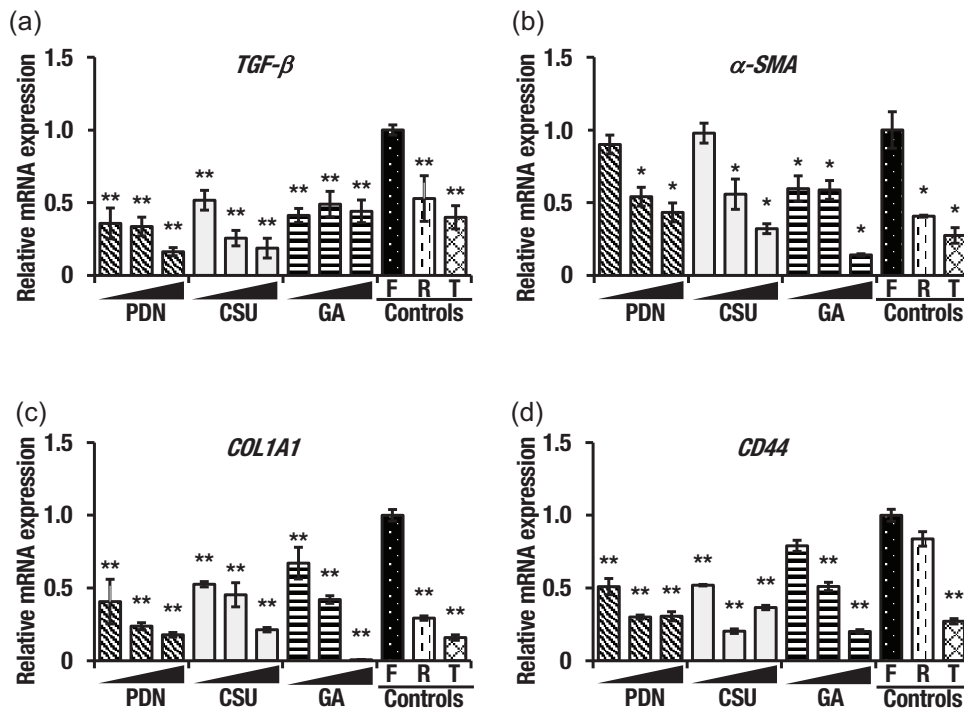


Figure 7. PDN, CSU, and GA mediate fibrosis via downregulating the mRNA expression levels of key fibrotic markers. (a–d) mRNA expression levels of key fibrotic markers, TGF- β (a), α -SMA (b), COL1A1 (c), and CD44 (d) were measured with RT-qPCR after treatment with different concentrations (25, 50, and 100 μ M) of different isolated compounds and controls (fibrosis induced group (control F); treated with 10% FBS only, gradually recovered group (control R); treated with 2% FBS and treatment positive group (control T); treated with 5 μ M of doxorubicin) for 24 h in activated LX-2 cells. Expression levels are normalized to 18S rRNA level. Data based on the value from control F set as 1.0. * P < 0.05, ** P < 0.01 (one way ANOVA with Dunnett's test). Each data is mean $\pm$ SD (n = 3).

Although a slow natural recovery is observed in the INT group, the damage caused by the TAA persists even after removal of the hepatotoxin. This is the critical stage that we need to overcome for the benefit of patients who already have established liver fibrosis. Interestingly, we observed a significant recovery in the treatment of CLS and BLE compared to the natural regeneration. This suggests that the extracts may accelerate the natural regeneration process compared to the natural recovery. This may be due to two reasons: either the leaf extract was involved in the desolation of the fibrous bridges, or the leaf extract prevented the formation of fibrous bridges from the start of the treatment. This is clear evidence establishing the fact that both extracts can effectively heal the liver. These observations of therapeutic recovery were similar to the results of the previously conducted prophylactic studies (Bogawaththa et al. 2021), suggesting that *O. octandra* has both prophylactic effect and therapeutic effect. These findings suggest that *O. octandra* has hepatoprotective, anti-inflammatory, and anti-fibrotic properties against TAA-induced liver fibrosis. On the other hand, previous studies have shown that the *O. octandra* plant contains a high amount of phenolic compounds, which are rich in antioxidant properties (Thabrew et al. 1995; Thabrew and Jayatilaka 1999; Perera, Ekanayake and Ranaweera 2015). Therefore, it can be hypothesized that the rapid recovery of chronic liver injury may be due to the antioxidants and other supportive compounds present in *O. octandra*. There is a lack of evidence on what kind of phenolic compounds are present in *O. octandra* and how they are intervened in attenuating or reversing of fibrosis.

In this study, we established an *in vitro* model to screen antifibrotic compounds. We focused on HSCs because they are cru-

cial and initial players in liver fibrosis. Normally, these cells are in a quiescent state, and they get activated in response to liver injury (Kamm and McCommis 2022). This activation is stimulated by factors such as TGF- β , which binds to specific receptors and triggers signaling pathways that lead to phenotypic changes and the transformation of HSCs into myofibroblast-like cells (Gutierrez-Ruiz and Gomez-Quiroz 2007). This activation increases the expression of α -SMA and stimulates the production and accumulation of collagen, ultimately contributing to the development of liver fibrosis (Hoffmann et al. 2020). Therefore, finding ways to intervene in the activation of HSCs is crucial to prevent or reverse liver fibrosis and protect the liver from further damage. Here, we used an immortalized human HSC cell line, LX-2, which allows for accessible in-depth mechanistic investigations without the use of cancer cell lines or animal models (Xu et al. 2005; Shu et al. 2021). Typically, these LX-2 are treated with CCL₄ or TGF- β to induce activation *in vitro*. LX-2 cells show quiescent state under 2% low serum conditions. In the presence of 10% high serum, LX-2 cells are activated and formed fibrotic features, making it an accessible model to study HSCs activation without the need for CCL₄ or TGF- β (Senoo et al. 2018; Schinagl et al. 2021).

Secondly, as for the first time, we were able to isolate the antifibrotic compounds, PDN, CSU and GA from the BLE of *O. octandra* using the LX-2 antifibrotic agent screening system. PDN and CSU are polyphenols belonging to the group of hydrolysable ellagitannins which generate ellagic acid by oxidative degradation (Landete 2011; Lee et al. 2022). Furthermore, this is the first time that PDN and CSU have been isolated from *O. octandra*. However, the effect of ellagitannin on liver fibrosis remains largely

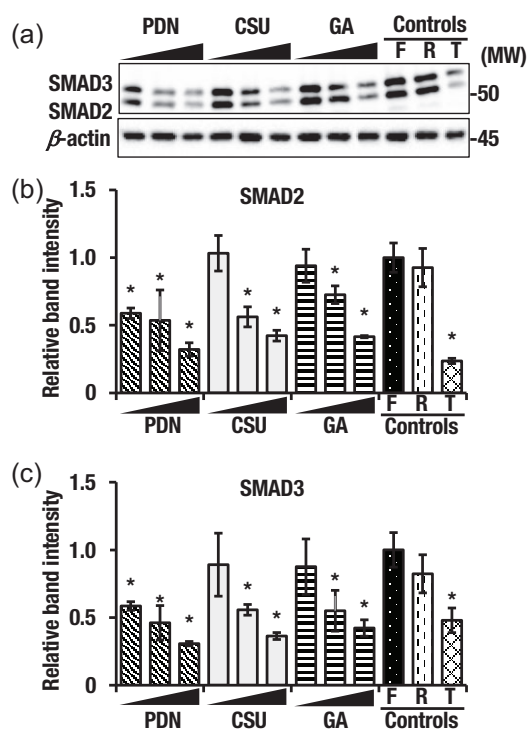


Figure 8. PDN, CSU and GA mediate fibrosis via regulation of SMAD protein expression. (a) Immunoblot analyses of SMAD2 and SMAD3: activated LX-2 cells were treated with different concentrations (25, 50, and 100 μ M) of different isolated compounds and controls (fibrosis induced group (control F); treated with 10% FBS only, gradually recovered group (control R); treated with 2% FBS and treatment positive group (control T); treated with 5 μ M of doxorubicin for 24 h. Equal loading was assed by probing the blots with antibody against β -actin. (b–c) Relative band intensity of SMAD2 (b) and SMAD3 (c). Band intensities were measured with image J software. Data based on the value from control F set as 1.0. * $P < 0.05$, ** $P < 0.01$ (one way ANOVA with Dunnett's test). Each data is mean $\pm$ SD ($n = 3$).

unknown. Although, GA has already been identified as a hepatoprotective agent (El-Lakkany *et al.* 2019), we compared the effect of PDN and CSU effect with GA. Interestingly, treatment with PDN and CSU reduced the transcription of α -SMA and COL1A1, showing significant intervention via the TGF- β signaling pathway. This observation may be due to the prevention of transcription of fibrotic genes and preventing the synthesis and deposition of ECM components in the liver (Goldsmith, Bradshaw and Spinale 2013). As a result, activated HSCs may prevent further activation or revert to their inactivated form suggesting the therapeutic potential. In addition, here to compare the anti fibrotic effect with natural recovery and doxorubicin which is one of the potential drug available in the market for liver fibrosis as mentioned above. Prominent antifibrosis effect could be observed for *O. octandra* suggesting significant therapeutic potential.

In this study, we investigated the expression of CD44, a newly identified fibrosis marker, during the treatment with isolated compounds. CD44 is expressed on the surface of HSCs and has been found to be upregulated during HSCs activation in liver fibrosis (Osawa *et al.* 2021). In addition, CD44 has been associated with HSCs survival and resistance to apoptosis (Li *et al.* 2020), which may contribute to the persistence of activated HSCs and the progression of liver fibrosis. Notably, PDN and CSU showed remarkable intervention on CD44 expression in activated LX-2 cells suggesting treatment potential to remove activated HSCs by transforming them into an inactivated form. Furthermore,

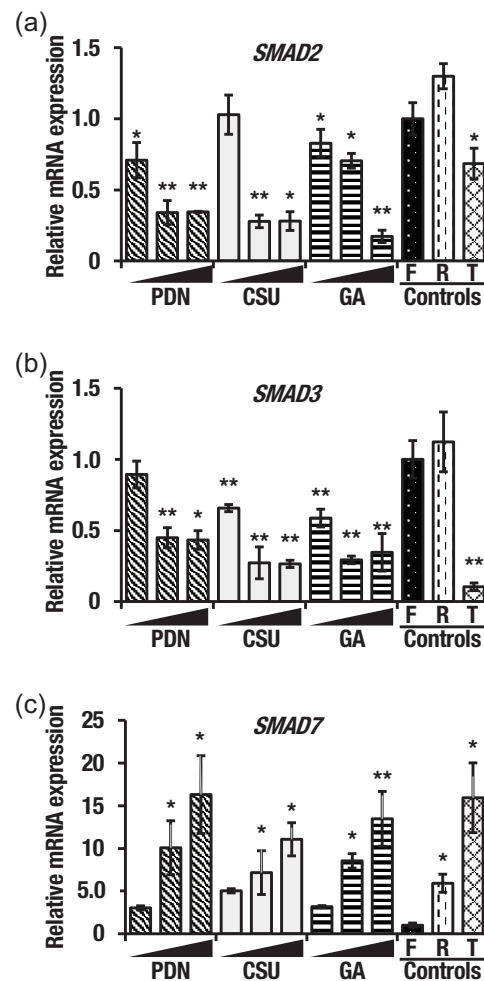


Figure 9. PDN, CSU, and GA mediate fibrosis via regulation of SMAD mRNA expression. (a–c) mRNA expression levels of SMAD2 (a), SMAD3 (b), and SMAD7 (c) were measured by RT-qPCR after treatment with various concentrations (25, 50, and 100 μ M) of different isolated compounds and controls (fibrosis induced group (control F); treated with 10% FBS only, gradually recovered group (control R); treated with 2% FBS and treatment positive group (control T); treated with 5 μ M of doxorubicin) for 24 h in activated LX-2 cells. Expression levels are normalized to 18S rRNA level. Data based on the value from control F set as 1.0. * $P < 0.05$, ** $P < 0.01$ (one way ANOVA with Dunnett's test). Each data is mean $\pm$ SD ($n = 3$).

PDN and CSU could accelerate the natural transformation of activated HSCs into their inactivated form by preventing the progression of fibrosis. These findings suggest that PDN and CSU may interfere with the activation of HSCs via the TGF- β signaling pathway.

As mentioned above, during the induction of fibrosis in HSCs, TGF- β induces conformational changes in SMAD2/3 in the cytoplasm, leading to its release from the receptor complex and binding to SMAD4 to form the SMAD2/3 complex, which then catalyze the phosphorylation of the serine residue on the downstream SMAD molecule (Souchevnytskyi *et al.* 2001). The phosphorylated SMAD then enters the nucleus and regulates the expression of TGF- β targets such as α -SMA and COL1A1, further inducing the onset of fibrosis (Yan *et al.* 2021). The SMAD family includes nine proteins, expressed as SMAD1-9, which are subdivided into receptor-activated SMAD (R-Smad), common pathway SMAD (Co-Smad), and inhibitory SMAD (I-Smad). R-Smad, such as SMAD2 and SMAD3, are mainly known to

promote liver fibrosis, whereas I-Smad, such as SMAD7 can inhibit or regulate the TGF- β family of signal transduction and inhibit liver fibrosis (Peng et al. 2022). Since the TGF- β /SMAD signaling pathway plays such an important role in liver fibrosis, blocking or mediating its signaling is an important potential strategy to prevent and treat liver fibrosis (Yuan et al. 2018). Interestingly, PDN, CSU, and GA downregulated SMAD2 and SMAD3 and upregulated SMAD7, thereby inhibiting the TGF- β /SMAD signaling pathway. In addition, our data indicate that natural gradual recovery of HSCs may require more time for recovery, suggesting that PDN and CSU have more potential to recover fibrosis than natural recovery and doxorubicin.

In this study, we focused only on three major compounds; PDN, CSU, and GA, which present in *O. octandra*, which were found in the dry leaves of *O. octandra*, at levels of 0.143%, 0.310%, and 0.012% respectively (data not shown). There may be another antifibrotic compound(s) responsible for the antifibrotic effect. Future experiments will be carried out to identify another possible compound(s) present in *O. octandra*. In addition, its therapeutic potential to interfere with fibrotic progression in kidney, lung, pancreas, and stomach and to reduce chronic inflammatory status will be investigated. In conclusion, the present study demonstrates the possible therapeutic potential of *O. octandra* to cure fibrosis via mediation of the TGF- β /SMAD signaling pathway and sheds light on the clinical significance for the other inflammatory diseases as well.

Acknowledgments

We acknowledge the technical support of Analytical Research Center for Experimental Sciences, Saga University. We would also like to thank Dr. Eranga H. Siriweera and the technical staff of the Department of Pathology Faculty of Medicine, University of Peradeniya for the IHC. Graphical abstract and Figure 4 were created with BioRender.com. This research was part of the dissertation submitted by the first author in partial fulfilment of a Ph.D. degree. All authors have given their consent.

Supplementary material

Supplementary material is available at *Bioscience, Biotechnology, and Biochemistry* online.

Data availability

The data underlying this article are available in the article and in its online supplementary material.

Author contribution

S.B.: conceptualization, methodology, investigation, formal analysis, investigation, writing—original draft. T.K.: formal analysis, investigation. U.B.: methodology, investigation, formal analysis, writing—original draft. T.H.: formal analysis, writing—original draft. G.Y.: formal analysis. H.I.: investigation, formal analysis. T.H.: investigation, formal analysis, writing—original draft. S.i.K.: investigation, formal analysis, supervision, writing—review & editing. K.K.W.: formal analysis, supervision. M.P.B.W.: conceptualization, supervision. K.I.: conceptualization, methodology, investigation, formal analysis, supervision, writing—review & editing. S.P.K.: conceptualization, methodology, supervision, writing—review & editing, funding acquisition. T.T.: conceptualization, methodology, supervision writing—review & editing, funding acquisition.

Funding

This study was funded by the [JSPS KAKENHI] under Grant [22H03515]; [Basis for Supporting Innovative Drug Discovery and Life Science Research (BINDS) from Japan Agency for Medical Research and Development (AMED)] under Grant [JP22ama121038j002]; [Advanced Research Infrastructure for Materials and Nanotechnology in Japan (ARIM) of the Ministry of Education, Culture, Sports, Science and Technology (MEXT)] under Grant [JPMXP1222KU1033] and [University of Peradeniya] under Grant [URG/2017/03/Ag].

Disclosure statement

The authors declare the following competing financial interests: S-i.K. and T.T. are advisory members of Tetra Create Co., Ltd. All other authors declare no conflicts of interest.

References

- Abdel-Rahman RF, Fayed HM, Asaad GF et al. The involvement of TGF- β 1/FAK/ α -SMA pathway in the antifibrotic impact of rice bran oil on thioacetamide-induced liver fibrosis in rats. *PLoS One* 2021;16:e0260130.
- Blanquicett C, Johnson MR, Heslin M et al. Housekeeping gene variability in normal and carcinomatous colorectal and liver tissues: applications in pharmacogenomic gene expression studies. *Anal Biochem* 2002;303:209–14.
- Bogahawaththa S, Kodithuwakku SP, Wijesundera KK et al. Antifibrotic and anti-angiogenic activities of *Osbeckia octandra* leaf extracts in thioacetamide-induced experimental liver cirrhosis. *Molecules* 2021;26:4836.
- Cao D, Shamsan E, Jiang B et al. Structural changes and expression of hepatic fibrosis-related proteins in coculture of *Echinococcus multilocularis* protoscoleces and human hepatic stellate cells. *Parasites Vectors* 2021;14:1–9.
- Chen D, Le TH, Shahidipour H et al. The role of gut-derived microbial antigens on liver fibrosis initiation and progression. *Cells* 2019;8:55–64.
- Dewidar B, Meyer C, Dooley S et al. Tgf- β in hepatic stellate cell activation and liver fibrogenesis—updated 2019. *Cells* 2019;8:1–35.
- Dwivedi DK, Jena GB. Glibenclamide protects against thioacetamide-induced hepatic damage in Wistar rat: investigation on NLRP3, MMP-2, and stellate cell activation. *Naunyn-Schmiedeberg's Arch Pharmacol* 2018;391:1257–74.
- El Awdan SA, Abdel Rahman RF, Ibrahim HM et al. Regression of fibrosis by cilostazol in a rat model of thioacetamide-induced liver fibrosis: up regulation of hepatic cAMP, and modulation of inflammatory, oxidative stress and apoptotic biomarkers. *PLoS One* 2019;14:e0216301.
- El-Lakkany NM, El-Maadawy WH, Seif el-Din SH et al. Antifibrotic effects of gallic acid on hepatic stellate cells: in vitro and in vivo mechanistic study. *J Tradit Complement Med* 2019;9: 45–53.
- Freedman ND, Curto TM, Morishima C et al. Silymarin use and liver disease progression in the Hepatitis C antiviral long-term treatment against cirrhosis trial. *Aliment Pharmacol Ther* 2011;33:127–37.
- Georges PC, Hui JJ, Gombos Z et al. Increased stiffness of the rat liver precedes matrix deposition: implications for fibrosis. *Am J Physiol Gastrointest Liver Physiol* 2007;293:1147–54.
- Golabi P, Paik JM, Eberly K et al. Causes of death in patients with non-alcoholic fatty liver disease (NAFLD), alcoholic liver

- disease and chronic viral Hepatitis B and C. *Ann Hepatol* 2022;**27**:100556.
- Goldsmith EC, Bradshaw AD, Spinale FG. Cellular mechanisms of tissue fibrosis. 2. Contributory pathways leading to myocardial fibrosis: moving beyond collagen expression. *Am J Physiol Cell Physiol* 2013;**304**:C393-402.
- Grayer RJ, Thabrew MI, Hughes RD et al. Phenolic and terpenoid constituents from the Sri Lankan medicinal plant *Osbeckia aspera*. *Pharm Biol* 2008;**46**:154-61.
- Gressner AM, Dooley S, Weiskirchen R. Tgf- β and the Smad pathway in liver fibrogenesis. *Signal Pathways Liver Diseases*. Berlin/Heidelberg: Springer-Verlag, 2005, 139-50.
- Greupink R, Bakker HI, Bouma W et al. The antiproliferative drug doxorubicin inhibits liver fibrosis in bile duct-ligated rats and can be selectively delivered to hepatic stellate cells in vivo. *J Pharmacol Exp Ther* 2006;**317**:514-21.
- Gutierrez-Ruiz MC, Gomez-Quiroz LE. Liver fibrosis: searching for cell model answers. *Liver Int* 2007;**27**:434-9.
- Haikal A, El-Neketi M, Elshaer S et al. *Mentha longifolia* subsp. typhoides and subsp. schimperii: antimicrobial and antiquorum-sensing bioactivities. *Chem Nat Compd* 2021;**57**:933-8.
- Hoffmann C, Djerir NEH, Danckaert A et al. Hepatic stellate cell hypertrophy is associated with metabolic liver fibrosis. *Sci Rep* 2020;**10**:1-13.
- Jaeschke H, Farhood A, Smith CW. Neutrophils contribute to ischemia/reperfusion injury in rat liver in vivo. *FASEB J* 1990;**4**:3355-9.
- Jantararussamee C, Rodniem S, Taweechoitipatr M et al. Hepatoprotective effect of probiotic lactic acid bacteria on thioacetamide-induced liver fibrosis in rats. *Probiotics Antimicro Prot* 2021;**13**:40-50.
- Kamm DR, McCommis KS. Hepatic stellate cells in physiology and pathology. *J Physiol* 2022;**600**:1825-37.
- Landete JM. Ellagitannins, ellagic acid and their derived metabolites: a review about source, metabolism, functions and health. *Food Res Int* 2011;**44**:1150-60.
- Lardizábal MN, Nocito AL, Daniele SM et al. Reference genes for real-time PCR quantification of microRNAs and messenger RNAs in rat models of hepatotoxicity. *PLoS One* 2012;**7**:e36323.
- Lavine JE, Schwimmer JB, Van Natta ML et al. Effect of vitamin E or metformin for treatment of nonalcoholic fatty liver disease in children and adolescents the tonic randomized controlled trial. *JAMA* 2011;**305**:1659-68.
- Lebeaupin C, Vallée D, Hazari Y et al. Endoplasmic reticulum stress signalling and the pathogenesis of non-alcoholic fatty liver disease. *J Hepatol* 2018;**69**:927-47.
- Lee GE, Kim RH, Lim T et al. Optimization of accelerated solvent extraction of ellagitannins in black raspberry seeds using artificial neural network coupled with genetic algorithm. *Food Chem* 2022;**396**:133712.
- Li W, Zhou C, Fu Y et al. Targeted delivery of hyaluronic acid nanomicelles to hepatic stellate cells in hepatic fibrosis rats. *Acta Pharmaceutica Sinica B* 2020;**10**:693-710.
- Murase K, Morrison KL, Tam PY et al. EF-Tu binding peptides identified, dissected, and affinity optimized by phage display. *Chem Biol* 2003;**10**:161-8.
- Ooki T, Hatakeyama M. Protocol for visualizing conditional interaction between transmembrane and cytoplasmic proteins. *STAR Protocols* 2021;**2**:100430.
- Osawa Y, Kawai H, Tsunoda T et al. Cluster of Differentiation 44 promotes liver fibrosis and serves as a biomarker in congestive hepatopathy. *Hepatol Commun* 2021;**5**:1437-47.
- Peng D, Fu M, Wang M et al. Targeting TGF- β signal transduction for fibrosis and cancer therapy. *Mol Cancer* 2022;**21**:1-20.
- Perera PR, Ekanayake S, Ranaweera KKD. Comparison of antiglycation and antioxidant potentials and total phenolic contents of decoctions from antidiabetic plants. *Procedia Chem* 2015;**16**:519-24.
- Prasadani M, Bogahawaththa S, Illeperuma RP et al. Leaf extract of *Osbeckia octandra* L. (Heen Bovitiya) suppresses human oral squamous cell carcinoma cells migration and induces cellular DNA damage. *J Oral Maxillofac Surg Med Pathol* 2021;**33**:215-20.
- Ramaiah SK. A toxicologist guide to the diagnostic interpretation of hepatic biochemical parameters. *Food Chem Toxicol* 2007;**45**:1551-7.
- Rhiouani H, El-Hilaly J, Israili Z et al. Acute and sub-chronic toxicity of an aqueous extract of the leaves of *Herniaria glabra* in rodents. *J Ethnopharmacol* 2008;**118**:378-86.
- Sahar S, Atrees HMR. Herbs and supplements for liver toxicity: a review on mode of action of herbs and supplements on liver toxicity. *Med J Cairo Univ* 2021;**89**:2179-83.
- Schinagl M, Tomin T, Gindlhuber J et al. Proteomic changes of activated hepatic stellate cells. *Int J Mol Sci* 2021;**22**:12782.
- Senoo T, Sasaki R, Akazawa Y et al. Geranylgeranylacetone attenuates fibrogenic activity and induces apoptosis in cultured human hepatic stellate cells and reduces liver fibrosis in carbon tetrachloride-treated mice. *BMC Gastroenterol* 2018;**18**:1-7.
- Shu Y, Liu X, Huang H et al. Research progress of natural compounds in anti-liver fibrosis by affecting autophagy of hepatic stellate cells. *Mol Biol Rep* 2021;**48**:1915-24.
- Singh A, Bajpai V, Kumar S et al. Profiling of gallic and ellagic acid derivatives in different plant parts of *Terminalia arjuna* by HPLC-ESI-QTOF-MS/MS. *Nat Prod Commun* 2016;**11**:239-44.
- Souchelnytskyi S, Rönnstrand L, Heldin CH et al. Phosphorylation of Smad signaling proteins by receptor serine/threonine kinases. *Methods Mol Biol* 2001;**124**:107-20.
- Thabrew M, Joice P, Rajatissa W. A comparative study of the efficacy of *Pavetta indica* and *Osbeckia octandra* in the treatment of liver dysfunction. *Planta Med* 1987;**53**:239-41.
- Thabrew MI, Gove CD, Hughes RD et al. Protective effects of *Osbeckia octandra* against galactosamine and tert-butyl hydroperoxide induced hepatocyte damage. *J Ethnopharmacol* 1995;**49**:69-76.
- Thabrew MI, Jayatilaka KAPW. A comparative study of the beneficial effects of *Osbeckia octandra* and *Osbeckia aspera* in liver dysfunction in rats. *Cey J Med Sci* 1999;**42**:1-4.
- Varga J, Pasche B. Antitransforming growth factor- β therapy in fibrosis: recent progress and implications for systemic sclerosis. *Curr Opin Rheumatol* 2008;**20**:720-8.
- Wang FS, Fan JG, Zhang Z et al. The global burden of liver disease: the major impact of China. *Hepatology* 2014;**60**:2099-108.
- Wijesundera KK, Izawa T, Tennakoon AH et al. M1-/M2-macrophages contribute to the development of GST-P-positive preneoplastic lesions in chemically-induced rat cirrhosis. *Exp Toxicol Pathol* 2015;**67**:467-75.
- Wong SW, Ting YW, Chan WK. Epidemiology of non-alcoholic fatty liver disease-related hepatocellular carcinoma and its implications. *JGH Open* 2018;**2**:235-41.
- Wong W-L, Abdulla MA, Chua K-H et al. Hepatoprotective effects of *Panus giganteus* (Berk.) corner against thioacetamide-(TAA-) induced liver injury in rats. *Evid Based Complement Alternat Med* 2012;**2012**:170303.

- Xu L, Hui AY, Albanis E et al. Human hepatic stellate cell lines, LX-1 and LX-2: new tools for analysis of hepatic fibrosis. *Gut* 2005;**54**:142-51.
- Yamanaka F, Hatano T, Ito H et al. Antibacterial effects of guava tannins and related polyphenols on *Vibrio* and *Aeromonas* species. *Nat Prod Commun* 2008;**3**:711-20.
- Yan J, Hu B, Shi W et al. Gli2 -regulated activation of hepatic stellate cells and liver fibrosis by TGF- β signaling. *Am J Physiol Liver Physiol* 2021;**320**:G720-8.
- Yang YM, Seki E. TNF α in liver fibrosis. *Curr Pathobiol Rep* 2015;**3**:253-61.
- Yuan X, Gong Z, Wang B et al. Astragaloside inhibits hepatic fibrosis by modulation of TGF- β 1/Smad signaling pathway. *Evid Based Complement Alternat Med* 2018;**2018**:3231647.
- Zhang A, Sun H, Wang X. Recent advances in natural products from plants for treatment of liver diseases. *Eur J Med Chem* 2013;**63**:570-7.
- Zhang QD, Xu MY, Cai XB et al. Myofibroblastic transformation of rat hepatic stellate cells: the role of Notch signaling and epithelial-mesenchymal transition regulation. *Eur Rev Med Pharmacol Sci* 2015;**19**:4130-8.

Received: 20 June 2023; Accepted: 4 August 2023

© The Author(s) 2023. Published by Oxford University Press on behalf of Japan Society for Bioscience, Biotechnology, and Agrochemistry. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com